Activity 1 - after.pdf

Improved prompt, corrected answer. NotebookLM response with sources P1-P5 selected (see prompt-design.txt for the prompt). Text reproduced from the returned response; only the layout (tables) and mathematical notation were reformatted for readability.

Key Concepts for Beginners

- **Recurrent Neural Network (RNN):** A type of neural network that processes text sequentially, one word at a time, carrying forward a running hidden state.
- **Self-Attention:** A mechanism that enables an algorithm to directly calculate mathematical relationships between all words across a text sequence simultaneously, irrespective of their physical distance.
- **BLEU (Bilingual Evaluation Understudy):** An automated scoring metric (typically scaled from 0 to 100) that evaluates translation quality by computing the overlap of word sequences (*n*-grams) between machine-generated output and human reference translations.
- **In-Context Learning:** The capability of a model to adapt to a task at test time strictly from prompt instructions and text demonstrations, without updating its neural network weights.
- **Few-Shot Learning:** Supplying a model with a natural language task description alongside a few completed input-output demonstration examples ($K \approx 10$–100) in its input prompt before prompting it for an answer.
- **Closed-Book QA:** A question-answering evaluation setting where the model must generate answers using only the factual knowledge stored internally within its parameters during pre-training, without searching external documents.
- **Exact Match (EM):** A strict evaluation metric for question answering that scores 1 if the generated answer string matches the human ground-truth reference character-for-character, and 0 otherwise.
- **Hallucination:** When a language model produces fluent, confident statements that are factually incorrect or ungrounded in real-world evidence.

P1: Attention Is All You Need (Vaswani et al., 2017; arXiv:1706.03762)

1. **Problem:** Dominant sequence transduction architectures rely on recurrent or convolutional neural networks that process tokens sequentially, which limits parallel computation during training and makes it computationally expensive to capture relationships between distant words (Section 1, pages 1–2; Section 4, page 6).
2. **Method, Data, and Evaluation Setting:**

- *Method:* The authors introduce the **Transformer**, an encoder-decoder architecture that dispenses entirely with recurrence and convolutions, relying solely on self-attention and multi-head attention to draw global dependencies between inputs and outputs (Section 1, page 2; Section 3.1–3.3, pages 2–5). The model connects all positions with a constant number of sequential operations and stacks $N=6$ identical layers in both the encoder and decoder, utilizing $h=8$ parallel attention heads in the base model and $h=16$ in the big model (Section 3.1–3.2, pages 2–5; Section 4, page 6; Table 3, page 9).
- *Data:* Trained on the standard WMT 2014 English-to-German dataset containing approximately 4.5 million sentence pairs (tokenized with byte-pair encoding into a 37,000-token shared vocabulary) and the WMT 2014 English-to-French dataset containing 36 million sentences (tokenized with word-piece into a 32,000-token vocabulary) (Section 5.1, page 7).
- *Evaluation Setting:* Evaluated on machine translation test sets (newstest2014) using beam search with a beam size of 4, length penalty $\alpha = 0.6$, and checkpoint averaging (averaging the last 5 checkpoints for base models and the last 20 checkpoints for big models) (Section 6.1, pages 7–8).

3. **Main Numerical Result:**

- *Metric:* BLEU score.
- *Dataset/Task:* WMT 2014 English-to-German translation on the newstest2014 test set (Section 6.1, page 8, Table 2).
- *Experimental Condition:* Transformer (big) model ($N=6$, $d_{\text{model}}=1024$, $h=16$) trained for 3.5 days (300,000 steps) on 8 NVIDIA P100 GPUs with an estimated computational cost of 2.3×10^{19} FLOPs (Section 6.1, page 8, Table 2; Table 3, page 9).
- *Reported Value:* **28.4 BLEU** (Section 6.1, page 8, Table 2).
- *Comparison Baseline:* Outperforms the best previously reported single-model and ensemble architectures, including the ConvS2S Ensemble (26.36 BLEU; training cost 7.7×10^{19} FLOPs) and the GNMT + RL Ensemble (26.30 BLEU; training cost 1.8×10^{20} FLOPs), by more than 2.0 BLEU (Section 6.1,

page 8, Table 2).

- *Result Selection Rationale:* I chose the WMT 2014 English-to-German result because the authors highlight it as the primary result establishing a new state of the art that surpasses all previous systems and ensembles by more than 2.0 BLEU at a fraction of the training compute (Section 6.1, page 8).

4. **Author-Stated Limitation:** The authors explicitly note that when sequence length *n* is very large (specifically when sequence length *n* exceeds representation dimensionality *d*), self-attention layers become computationally slower than recurrent layers, which would require restricting self-attention to only consider a local neighborhood of size *r* centered around each position (increasing the maximum path length to O(*n*/*r*)) (Section 4, page 6, Table 1).

P2: Language Models are Few-Shot Learners (Brown et al., 2020; arXiv:2005.14165)

1. **Problem:** Standard natural language processing systems require task-specific fine-tuning on large supervised datasets of thousands to hundreds of thousands of examples, which limits applicability to new tasks, risks exploiting spurious dataset correlations, and fails to match human-like sample efficiency (Section 1, pages 3–4).
2. **Method, Data, and Evaluation Setting:**

- *Method:* The authors scale autoregressive Transformer decoders up to 175 billion parameters (**GPT-3**), utilizing alternating dense and locally banded sparse attention patterns across 96 layers and 96 attention heads (Section 2.1, page 8, Table 2.1). GPT-3 is applied entirely without fine-tuning or gradient updates (Section 1, page 4).
- *Data:* Pre-trained on 300 billion tokens comprising filtered Common Crawl (weighted 60%), WebText2 (22%), Books1 (8%), Books2 (8%), and Wikipedia (3%) (Section 2.2, pages 8–9, Table 2.1; Table 2.2).
- *Evaluation Setting:* Evaluated on over two dozen NLP benchmarks across three prompt-based conditions: zero-shot (natural language task description only), one-shot (task description plus 1 demonstration), and few-shot (task description plus *K* demonstrations, typically 10 to 100 fitting the 2,048-token context window) using feed-forward inference (Section 1, page 4; Section 2, page 7; Figure 2.1, page 6).

3. **Main Numerical Result:**

- *Metric:* Exact accuracy (%).
- *Dataset/Task:* TriviaQA question answering evaluated in the closed-book setting (Section 3.2, pages 12–13, Table 3.3).
- *Experimental Condition:* GPT-3 175B in the few-shot setting with *K*=64 in-context demonstrations, evaluated on the test server without fine-tuning or external document retrieval (Section 3.2, Table 3.3, page 13; Figure 3.3, page 14; Appendix H, Table H, page 63).
- *Reported Value:* **71.2% accuracy** (Section 3.2, Table 3.3, page 13).
- *Comparison Baseline:* Outperforms the state-of-the-art fine-tuned closed-book baseline T5-11B+SSM (60.5% accuracy) by 10.7 percentage points, and exceeds the fine-tuned open-domain retrieval system RAG (68.0% accuracy) (Section 3.2, Table 3.3, page 13; Figure 3.3, page 14).
- *Result Selection Rationale:* I chose the TriviaQA closed-book result because the authors highlight it as the benchmark where GPT-3's few-shot capability not only outperforms fine-tuned closed-book systems but also surpasses a specialized open-domain system equipped with document retrieval (Section 3.2, page 13; Figure 3.3, page 14).

4. **Author-Stated Limitation:** The authors explicitly report that GPT-3's strictly autoregressive (left-to-right) architecture creates structural limitations on comparison tasks that require looking back and comparing two pieces of content or bidirectionality—such as Word-in-Context (WiC, scoring 49.4% accuracy, near chance) and Adversarial Natural Language Inference (ANLI Round 3, scoring 40.2% accuracy)—as well as on several reading comprehension benchmarks (Section 3.7, Table 3.8, page 19; Section 5, page 33).

P3: Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks (Lewis et al., 2020; arXiv:2005.11401)

1. **Problem:** Large pre-trained language models store factual knowledge implicitly within their network parameters, making it difficult to expand or revise their memory, inspect decision provenance, or prevent hallucinations on knowledge-intensive tasks (Section 1, pages 1–2).
2. **Method, Data, and Evaluation Setting:**

- *Method:* The authors develop **Retrieval-Augmented Generation (RAG)**, combining parametric memory (a pre-trained BART-large sequence-to-sequence model with ~400M parameters) with

non-parametric memory (a dense vector index of Wikipedia) accessed via a neural retriever (Dense Passage Retriever / DPR, based on BERT-base bi-encoders) (Section 1, page 2; Section 2.1–2.2, pages 2–3). The system is trained end-to-end to marginalize over latent retrieved documents via two formulations: RAG-Sequence (conditioning on the same retrieved passages across the whole sequence) and RAG-Token (conditioning on different retrieved passages per generated token) (Section 2.1, pages 2–3).

- *Data:* Non-parametric memory consists of a December 2018 Wikipedia dump split into 21 million disjoint 100-word chunks indexed using FAISS Maximum Inner Product Search (MIPS) (Section 3, page 4). Evaluated on open-domain QA (Natural Questions, TriviaQA, WebQuestions, CuratedTrec), abstractive QA (MS-MARCO), Jeopardy question generation, and fact verification (FEVER) (Section 3.1–3.4, pages 4–5).
- *Evaluation Setting:* Models are fine-tuned on input-output pairs; for open-domain QA, the retriever accesses $k \in \{5, 10\}$ documents during training and evaluates with 15 documents (RAG-Token) or 50 documents (RAG-Sequence) using greedy decoding, evaluated with Exact Match (EM) scores (Section 3, page 4; Appendix A, page 17).

3. **Main Numerical Result:**

- *Metric:* Exact Match (EM) score.
- *Dataset/Task:* Open-Domain Natural Questions (NQ) test set (Section 4.1, Table 1, page 6).
- *Experimental Condition:* RAG-Sequence fine-tuned end-to-end and evaluated with 50 retrieved documents using thorough decoding (Section 4.1, Table 1, page 6; Appendix A, page 17).
- *Reported Value:* **44.5 EM** (Section 4.1, Table 1, page 6).
- *Comparison Baseline:* Outperforms the closed-book parametric baseline T5-11B+SSM (36.6 EM) by 7.9 points, and outperforms open-book retrieve-and-extract architectures REALM (40.4 EM) and DPR (41.5 EM) (Section 4.1, Table 1, page 6).
- *Result Selection Rationale:* I chose the Natural Questions result because the authors highlight it as the primary open-domain benchmark proving that an unconstrained generative model with non-parametric retrieval outperforms specialized extractive reader-ranker systems (Section 4.1, pages 5–6, Table 1).

4. **Author-Stated Limitation:** The authors explicitly report a failure mode called "retrieval collapse" observed in preliminary experiments on tasks like story generation, where the retrieval component would collapse and learn to retrieve the exact same documents regardless of input, leading the generator to completely ignore retrieved documents and revert to parametric BART behavior (Appendix H, page 19).

P4: ReAct: Synergizing Reasoning and Acting in Language Models (Yao et al., 2022; arXiv:2210.03629)

1. **Problem:** Large language models traditionally isolate reasoning (such as internal Chain-of-Thought) and interactive action planning, causing reasoning systems to suffer from ungrounded factual hallucinations and error propagation, while action-only systems lack working memory and high-level goal decomposition (Section 1, pages 1–2).

2. **Method, Data, and Evaluation Setting:**

- *Method:* The authors propose **ReAct**, a prompting paradigm that expands an agent's action space to include free-form language thoughts ($\hat{A} = A \cup L$), interleaving verbal reasoning traces and environment actions so that reasoning dynamically guides and updates action plans while environment observations inform further reasoning (Section 1, page 2; Section 2, pages 2–3).
- *Data and Benchmarks:* Evaluated on multi-hop question answering (HotpotQA, question-only setup), fact verification (FEVER), simulated household interaction (ALFWorld, 134 unseen evaluation games across 6 task types), and web shopping (WebShop, 1.18M products, 500 test instructions) (Section 3.1, page 4; Section 4, pages 6–8).
- *Evaluation Setting:* Evaluated primarily with a frozen PaLM-540B language model prompted with 1 to 6 in-context few-shot human trajectories under greedy decoding without gradient updates (Section 2, page 3; Section 3.2, page 4; Section 4, pages 6–8).

3. **Main Numerical Result:**

- *Metric:* Task Success Rate (SR, %).
- *Dataset/Task:* ALFWorld interactive household text environment on 134 unseen evaluation games (Section 4, page 8, Table 3).
- *Experimental Condition:* ReAct prompting using PaLM-540B with 2 in-context example trajectories (best trial out of 6 prompt permutations) (Section 4, page 8, Table 3).

- *Reported Value:* **71% success rate** (best of 6; average across all 6 trials was 57%) (Section 4, page 8, Table 3).
- *Comparison Baseline:* Substantially outperforms the imitation learning baseline BUTLER (best of 8 trials: 37% success rate) trained on 10^5 expert trajectories, and the Act-only prompting baseline (best of 6 trials: 45% success rate) (Section 4, page 8, Table 3).
- *Result Selection Rationale:* I chose the ALFWorld interactive decision-making result because the authors highlight it as the most dramatic demonstration of ReAct's synergy, nearly doubling the performance of an imitation learning agent trained on 100,000 domain demonstrations using only 2 prompt examples (Section 1, page 1; Section 4, page 8).

4. **Author-Stated Limitation:** The authors explicitly report that ReAct's strict structure of interleaving steps reduces reasoning flexibility, resulting in a higher reasoning error rate than Chain-of-Thought (47% vs. 16% on HotpotQA), including a frequent failure mode where the model gets trapped in repetitive loops generating identical thoughts and actions without recovering (Section 3.3, pages 5–6, Table 2).

P5: Self-Refine: Iterative Refinement with Self-Feedback (Madaan et al., 2023; arXiv:2303.17651)

1. **Problem:** Large language models often fail to satisfy multifaceted or complex generation requirements on their first attempt, whereas existing refinement systems rely on expensive supervised training data, human annotations, or external reward models (Section 1, page 1).
2. **Method, Data, and Evaluation Setting:**

- *Method:* The authors present **SELF-REFINE**, a training-free framework where a single pre-trained language model serves as generator, feedback provider, and refiner; it produces an initial output y_0, uses few-shot prompts to generate multi-aspect natural language feedback fb_t, and iteratively refines its output into y_{t+1} while maintaining a history of past feedback and attempts (Section 1, page 1; Section 2, pages 2–3).
- *Data:* Evaluated across 7 diverse tasks: Dialogue Response Generation on the FED dataset (372 conversations [Table 4, page 14]; 342 test instances used for automatic evaluation and 100 instances evaluated by humans [Appendix M, page 33]), Code Optimization (PIE dataset), Code Readability Improvement (CodeNet, 300 snippets), Math Reasoning (GSM-8k), Sentiment Reversal (Yelp reviews, 1,000 passages), Acronym Generation (250 acronyms), and Constrained Generation (CommonGen-Hard with 20–30 concepts) (Section 3, page 4; Table 4, page 14; Appendices L–R).
- *Evaluation Setting:* Evaluated using frozen models (GPT-3.5 `text-davinci-003`, ChatGPT `gpt-3.5-turbo`, GPT-4, and CODEX `code-davinci-002`) running with temperature 0.7 across up to 4 iterations (specifically a maximum of k=3 iterations for dialogue response generation [Appendix M.2, page 32]), comparing refined outputs directly against the base model's unrefined initial output (Section 3.1–3.2, pages 4–5).

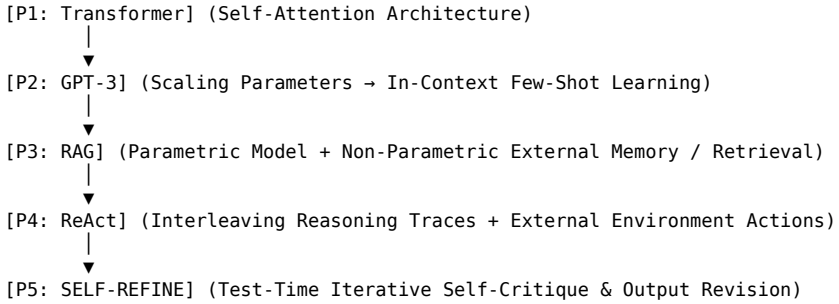
3. **Main Numerical Result:**

- *Metric:* Preference score (%) (the judge is uncertain because the source passages conflict: Section 3.2 on page 5 mentions GPT-4 as a proxy for human preference, whereas Appendix M on page 33 states that the automatic evaluation judge was text-davinci-003).
- *Dataset/Task:* Dialogue Response Generation on the FED dataset (Section 3.3, Table 1, page 5; Section 3.2, page 5; Appendix M.2, page 32; Appendix M, page 33).
- *Experimental Condition:* Base GPT-4 evaluated with SELF-REFINE using a maximum of k=3 iterations compared directly against Base GPT-4 one-step generation (Section 3.1–3.3, Table 1, page 5; Appendix M.2, page 32).
- *Reported Value:* **74.6% preference score** for GPT-4 + SELF-REFINE versus 25.4% for Base GPT-4, representing an absolute improvement of **+49.2 percentage points** (Section 3.3, Table 1, page 5; Section 3.3, page 5).
- *Comparison Baseline:* Base GPT-4 direct one-step generation without feedback-refinement iterations (25.4% preference score) (Section 3.3, Table 1, page 5).
- *Result Selection Rationale:* I chose the Dialogue Response Generation result on GPT-4 because it exhibits the largest absolute performance improvement (+49.2 percentage points) among all evaluated benchmarks in the paper, illustrating the impact of iterative self-critique on multifaceted open-ended tasks (Section 3.3, Table 1, page 5; Section 3.3, page 5).

4. **Author-Stated Limitation:** The authors explicitly state as their primary limitation that the underlying base model must already possess sufficient few-shot or instruction-following capabilities to generate structured feedback and execute edits, noting that weaker models like Vicuna-13B fail to produce formatted feedback, repeat identical text, or hallucinate (Section 4, pages 7–8; Section 6, page 9).

Comparison of the Five Approaches

These five papers trace how language models evolved from architectural foundations to interactive, test-time problem solvers:



Attention Is All You Need (P1) establishes the parallelizable self-attention backbone (the Transformer) that replaces recurrent bottlenecks. **GPT-3 (P2)** scales this architecture to 175 billion parameters, demonstrating that enormous Transformer decoders can perform tasks few-shot via in-context learning without updating network weights. However, GPT-3 relies on static parameter weights, creating vulnerability to factual hallucinations and outdated facts. **RAG (P3)** addresses this parameter ceiling by pairing sequence-to-sequence generators with external Wikipedia retrieval, ground-truthing generations in non-parametric text. While RAG retrieves passively, **ReAct (P4)** turns models into active agents, interleaving verbal reasoning traces with environment actions to plan, search, and recover dynamically. Finally, **SELF-REFINE (P5)** operates purely at test time on the output text itself, closing the loop by prompting a single model to critique and iteratively revise its own generated drafts without external retrievers or retraining.

Summary Comparison Table

Paper ID & Citation	Core Mechanism	Primary Dataset / Task	Main Numerical Result	Baseline Comparison	Stated Limitation
P1 (Vaswani et al., 2017)	Multi-Head Self-Attention Transformer	WMT 2014 English-to-German	**28.4 BLEU** (Transformer big)	ConvS2S Ensemble (26.36 BLEU); GNMT+RL Ensemble (26.30 BLEU)	Slower than RNNs when sequence length $n >$ d_{model}
P2 (Brown et al., 2020)	175B Autoregressive Few-Shot Scaling	TriviaQA (Closed-Book QA)	**71.2% accuracy** (K=64 demonstrations)	Fine-tuned T5-11B+SSM (60.5%); Fine-tuned RAG (68.0%)	Struggles on compariso n/bidirectional tasks (WiC, ANLI)
P3 (Lewis et al., 2020)	Hybrid Parametric + Non-Parametric Retrieval	Natural Questions (Open-Domain QA)	**44.5 EM** (RAG-Sequence)	Closed-book T5-11B+SSM (36.6 EM); Extractive DPR (41.5 EM)	Retrieval collapse on tasks with less explicit factual demands (Appendix H, page 19)
P4 (Yao et al., 2022)	Interleaving Reasoning Traces & External Actions	ALFWorld (Simulated Household Game)	**71% success rate** (best of 6 trials)	Imitation Learning BUTLER (37%); Act-only prompt (45%)	Reduced reasoning flexibility; repetitive loops in thoughts/actions
P5 (Madaan et al., 2023)	Iterative Self-Feedback & Output Refinement	FED (Dialogue Response Generation)	**74.6% preference score** (GPT-4 + SELF-REFINE, k=3; judge uncertain)	Base GPT-4 direct one-step generation (25.4% preference)	Requires strong base LLMs; fails on smaller models like Vicuna-13B

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paper_id,problem,method,data,evaluation_setting,metric,result,baseline_if_reported,limitation,source_location
P1,"Dominant sequence transduction models rely on complex recurrent or convolutional neural networks that process tokens sequentially, limiting parallelization during training and making the modeling of dependencies between distant positions computationally expensive.","The Transformer architecture relies entirely on multi-head self-attention mechanisms (6 encoder and 6 decoder layers with 8 parallel heads in base model and 16 heads in big model) to model global dependencies without recurrence or convolutions.","WMT 2014 English-to-German dataset (approximately 4.5 million sentence pairs) and WMT 2014 English-to-French dataset (36 million sentences).","Evaluation on newstest2014 using beam search with beam size 4 and length penalty alpha = 0.6, with checkpoint averaging of the last 20 checkpoints for the big model.","BLEU score","28.4 BLEU","ConvS2S Ensemble (26.36 BLEU) and GNMT + RL Ensemble (26.30 BLEU)","Computational complexity per layer for self-attention is O(n^2 * d), becoming slower than recurrent layers when sequence length n exceeds representation dimensionality d.","Method: Section 3.1-3.3 (pages 2-5; Table 3, page 9); Result: Section 6.1 (page 8, Table 2)"
P2,"Traditional NLP systems require task-specific fine-tuning on large supervised datasets of thousands to tens of thousands of examples, limiting generalization and adaptability to new tasks.","Scaling an autoregressive Transformer decoder to 175 billion parameters (GPT-3) across 96 layers and 96 attention heads with alternating dense and locally banded sparse attention, evaluated purely via in-context prompt learning without fine-tuning or gradient updates.","Pre-trained on 300 billion tokens comprising filtered Common Crawl (60%), WebText2 (22%), Books1 (8%), Books2 (8%), and Wikipedia (3%).","Closed-book TriviaQA question answering evaluated in the few-shot setting with K = 64 in-context demonstrations on the wiki split test server without external retrieval.","Accuracy (%)","71.2%","Fine-tuned closed-book T5-11B+SSM (60.5%) and fine-tuned open-domain RAG (68.0%)","Autoregressive left-to-right architecture lacks bidirectionality, resulting in poor performance on sentence comparison tasks like WiC (49.4% accuracy) and ANLI Round 3 (40.2% accuracy).","Method: Section 2.1-2.2 (pages 8-9, Table 2.1); Result: Section 3.2 (page 13, Table 3.3)"
P3,"Large pre-trained language models store factual knowledge implicitly in parameters, making knowledge expansion, updating, and provenance inspection difficult while risking factual hallucinations.","Retrieval-Augmented Generation (RAG) combining parametric memory (BART-large, 400M parameters) with non-parametric memory (a dense vector index of Wikipedia accessed via DPR BERT-base bi-encoders) and marginalizing over latent retrieved documents.","December 2018 Wikipedia dump split into 21 million disjoint 100-word chunks indexed using FAISS MIPS.","Open-domain Natural Questions test set evaluated with RAG-Sequence using 50 retrieved documents and thorough decoding.","Exact Match (EM)","44.5 EM","Closed-book T5-11B+SSM (36.6 EM), open-book REALM (40.4 EM), and DPR (41.5 EM)","Susceptibility to retrieval collapse on tasks with less explicit factual demands (e.g., story generation), where the retriever repeatedly returns
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identical passages and the generator learns to ignore them.","Method: Section 2.1-2.3 (pages 2-3); Result: Section 4.1 (page 6, Table 1); Limitation: Appendix H (page 19)"

P4,"Language models traditionally decouple reasoning and acting, causing internal reasoning traces to suffer from ungrounded hallucinations while action-only policies lack goal decomposition and working memory.","ReAct prompting paradigm expanding the action space to interleave verbal reasoning thoughts with task-specific environment actions, enabling dynamic planning, execution, and observation tracking.","134 unseen evaluation games across 6 task types in the simulated household ALFWorld text environment.","PaLM-540B evaluated with 2-shot in-context trajectories using greedy decoding (best trial of 6 prompt permutations).","Success Rate (%)","71%","Imitation learning baseline BUTLER (37% success rate) trained on 10^5 trajectories, and Act-only prompting (45% success rate)","Interleaving thoughts and actions reduces reasoning flexibility, increasing reasoning errors over CoT (47% vs. 16% on HotpotQA) and creating failure loops of repetitive actions and thoughts.","Method: Section 2 (pages 2-3); Result: Section 4 (page 8, Table 3)"

P5,"Language models often fail to meet complex generation requirements on their first attempt, while existing refinement methods depend on expensive supervised training data or external reward models.","SELF-REFINE training-free iterative loop where a single language model acts as generator, feedback provider, and refiner using few-shot multi-aspect feedback prompts across up to 4 iterations (maximum of k=3 iterations for dialogue response generation).","FED dataset for Dialogue Response Generation (372 conversations; 342 test instances used for automatic evaluation and 100 instances evaluated by humans).","Frozen GPT-4 evaluated on dialogue response generation with a maximum of k=3 iterations and temperature 0.7 (judge is uncertain due to conflicting descriptions: Section 3.2 on page 5 cites GPT-4, while Appendix M on page 33 cites text-davinci-003).","Preference Score (%)","74.6%","Base GPT-4 direct one-step generation (25.4% preference score)","Depends heavily on strong base model capabilities; weaker models like Vicuna-13B fail to produce formatted multi-aspect feedback or achieve meaningful output improvements.","Method: Section 2 (pages 2-3); Result: Section 3.3 (page 5, Table 1); Settings & Data: Table 4 (page 14), Appendix M.2 (page 32), Appendix M (page 33)"

Verification Guide Table

Paper ID	Paper Title	Method Claim to Verify	Method Location	Result Claim to Verify	Result Location
P1	*Attention Is All You Need*	Transformer architecture relies purely on self-attention without recurrence; big configuration specifies d_{model}=1024, h=16.	**Section 3.1–3.3 (pages 2–5); Table 3 (page 9)**	Transformer (big) reaches **28.4 BLEU** on WMT'14 En-De, outperforming previous ensembles.	**Section 6.1 (page 8, Table 2)**
P2	*Language Models are Few-Shot Learners*	GPT-3 scales autoregressive decoding to 175B parameters across 96 layers and 96 heads without gradient updates.	**Section 2.1–2.2 (pages 8–9, Table 2.1)**	Closed-book GPT-3 175B achieves **71.2% accuracy** on TriviaQA in the few-shot (K=64) setting.	**Section 3.2 (page 13, Table 3.3)**
P3	*Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*	RAG combines parametric BART-large with non-parametric DPR retrieval over 21M Wikipedia passages.	**Section 2.1–2.3 (pages 2–3, Figure 1)**	RAG-Sequence achieves **44.5 EM** on the Natural Questions test set.	**Section 4.1 (page 6, Table 1)**
P4	*ReAct: Synergizing Reasoning and Acting in Language Models*	Action space is augmented with language thoughts ($\hat{A} = A \cup L$) to interleave reasoning and acting.	**Section 2 (pages 2–3, Figure 1)**	PaLM-540B ReAct achieves a **71% success rate** on 134 unseen ALFWorld games (best of 6 trials).	**Section 4 (page 8, Table 3)**
P5	*Self-Refine: Iterative Refinement with Self-Feedback*	A single frozen model iteratively generates, critiques, and refines outputs across multi-aspect rubrics.	**Section 2 (pages 2–3, Figure 1)**	GPT-4 + SELF-REFINE attains a **74.6% preference score** on dialogue response generation (k=3; judge uncertain).	**Section 3.3 (page 5, Table 1); Appendix M (pages 32–33)**

Would you like to explore how an engineering student might write an end-to-end evaluation script to benchmark these five architectures on a unified reasoning task?